

STOR 155.010 Lecture 07

Linear Regression I

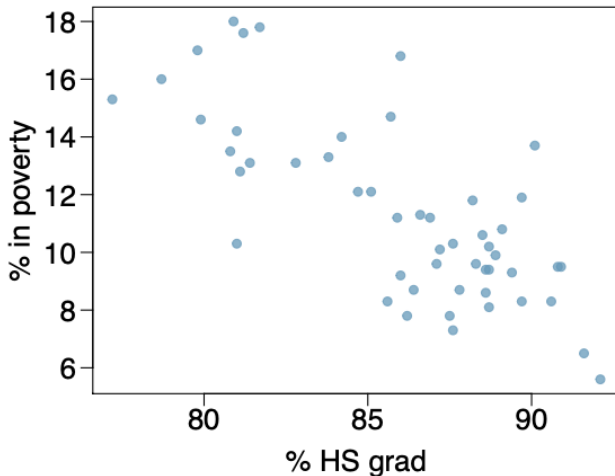
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Poverty vs high school graduation rate

Relationship between HS graduate rate in all 50 US states and DC and the % of residents who live below the poverty line:



Model fitting

- The simplest theoretical model to fit on the data is a **linear model**.
- The best linear model for predicting poverty from high school graduation rate in the US is

$$\hat{poverty} = 64.78 - 0.62 \cdot HS_{grad}$$

The “hat” is used to signify that this is an estimate.

- The high school graduate rate in Georgia is 85.1%. What poverty level does the model predict for this state?

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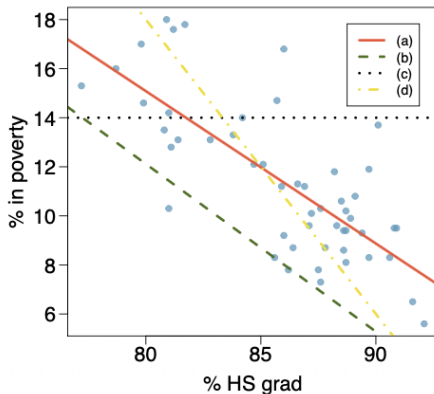
- The high school graduate rate in Georgia is 85.1%. What poverty level does the model predict for this state?

$$64.78 - 0.62 \cdot 85.1 = 12.018$$

- **This ability for prediction showcases the power of having a statistical model!**

Eyeballing the line

Which line best fits the relationship between % in poverty and % HS grad?

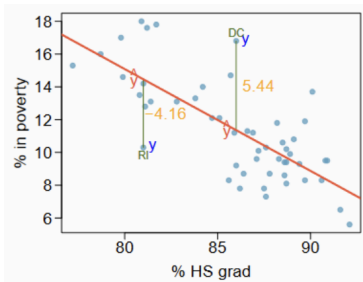
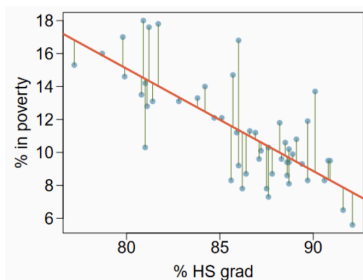


What is it that makes it the best line?

Residuals

Residuals are the leftovers from the fitted model

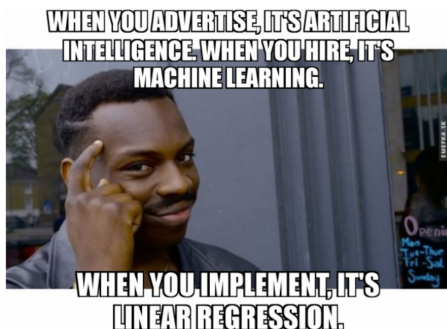
$$\text{Data} \approx \text{Fitted value} + \text{Residual}$$



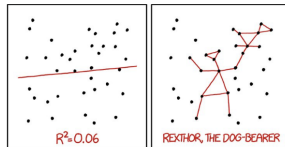
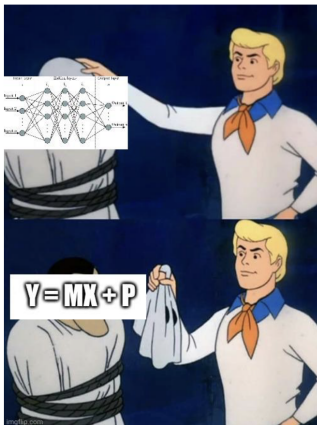
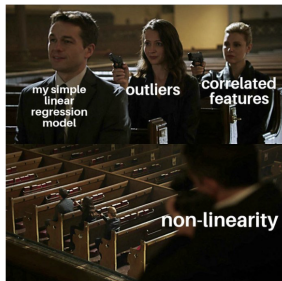
- % living in poverty in DC is 5.44% more than predicted.
- % living in poverty in RI is 4.16% less than predicted.

The “best” line

- Collectively small residuals.
- Minimize $\sum |\text{residuals}|$ or $\sum \text{residuals}^2$.
- The latter leads to **least squares regression**!
- The line that minimizes the sum of squares of residuals is called the *line of best fit*.

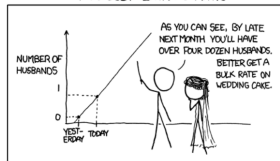


Linear Regression Explainers



I DON'T TRUST LINEAR REGRESSIONS WHEN IT'S HARDER TO GUESS THE DIRECTION OF THE CORRELATION FROM THE SCATTER PLOT THAN TO FIND NEW CONSTELLATIONS ON IT.

MY HOBBY: EXTRAPOLATING



Problem Setup

- Given pairs of data points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$.
- We want to construct a model $y_i = f(x_i) + \varepsilon_i$ that connects the variables X and Y .
- We focus on linear functions f :

$$y = f(x) = b_0 + b_1x.$$

- Such a model allows us to predict Y for any value of X .
- X is called the **explanatory** or **independent** or **predictor** variable.
- Y is called the **response** or **outcome** or **dependent** variable.

Finding the line of best fit

- Sum of squares of residuals $SSE = \sum_{i=1}^n (y_i - (b_0 + b_1 x_i))^2$. We want to minimize SSE.
- The best fitting (least squares) line is $\hat{y} = \hat{b}_0 + \hat{b}_1 x$, where

$$\arg \min_{b_0, b_1} \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2.$$

- Technical jargon:
 - $\hat{y}$ is the predicted value of the response variable.
 - $\hat{b}_0$ is the y-intercept.
 - $\hat{b}_1$ is the slope.
 - x is the value of the explanatory variable.

Finding the slope and intercept

- Let $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ and $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$. Also recall $s_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$ and $s_y^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2$.

- Slope:**

$$\hat{b}_1 = r \frac{s_y}{s_x} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

- Intercept:**

$$\hat{b}_0 = \bar{y} - \hat{b}_1 \bar{x}$$

- Once $\hat{b}_0$ and $\hat{b}_1$ are known, any **prediction** for y given a value of x can be made:

$$\hat{y} = \hat{b}_0 + \hat{b}_1 x.$$

Coefficient of Determination R^2

- Connection with Pearson's correlation.
- Informally speaking,

$$R^2 \approx \frac{\text{variation explained by the model}}{\text{total variation}}$$

- Example: If a linear model has a very strong negative relationship with a correlation of -0.97 , how much of the variation in the response is explained by the explanatory variable?

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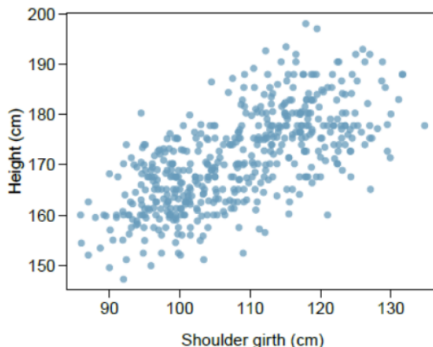
- Example: If a linear model has a very strong negative relationship with a correlation of -0.97 , how much of the variation in the response is explained by the explanatory variable?
- About $R^2 = (-0.97)^2 = 0.94$ or 94% of the variation is explained by the linear model.

Interpreting the model

- The **slope** describes the estimated difference in the y variable if the explanatory variable x for a case happened to be one unit larger.
- The **intercept** describes the average outcome of y if $x = 0$ and the linear model is valid all the way to $x = 0$, which in many applications is not the case.
- The resulting model allows us to give a more refined description of the relationship between variables x and y .
- However, no causality can be derived. **Correlation does not imply causation!**

Practice Example

Data on shoulder girth and height of adults. The mean shoulder girth is 107.60 cm with a standard deviation of 10.39 cm. The mean height is 171.15 cm with a standard deviation of 9.42 cm. The correlation between height and shoulder girth is 0.68.



Questions

- 1 Write the equation of the regression line for predicting height. (Let y represent height in centimeters and let x represent shoulder girth in centimeters. Round your numerical values to three decimal places.)
- 2 Interpret the slope and intercept in this context.
- 3 Calculate R^2 of the regression line for predicting height from shoulder girth. Interpret it in the context of the problem.
- 4 A randomly selected student from your class has a shoulder girth of 100 cm. Predict the height (in cm) of this student using the model.
- 5 The student from part (d) is 160 cm tall. Calculate the residual (in cm). Did the model underestimate or overestimate the student's height?

- ① From the given information, we see that

$$\bar{x} = 107.60, \quad s_x = 10.39$$

$$\bar{y} = 171.15, \quad s_y = 9.42$$

$$r = 0.68$$

Hence, we get the regression estimates

$$\hat{b}_1 = r \frac{s_y}{s_x} = 0.68 \cdot \frac{9.42}{10.39} = 0.617$$

$$\hat{b}_0 = \bar{y} - \hat{b}_1 \bar{x} = 171.15 - 0.617 \cdot 107.60 = 104.813$$

and the equation for the regression line becomes

$$\hat{y} = 0.617x + 104.813$$

Solutions (contd.)

- ② *Slope:* For each centimeter increase in shoulder girth, we would expect height to increase on average by 0.617 centimeters.

Intercept: People who have a shoulder girth of 0 centimeters are expected to be on average 104.813 centimeters tall.

③ $R^2 = (0.68)^2 = 0.4624$

Interpretation: Approximately 46.24% of the variation in heights is accounted for by the model.

Solutions (contd.)

- 4 The predicted height is

$$\hat{y} = 0.617 \cdot 100 + 104.813 = 166.46 \text{ cm.}$$

- 5 The residual in general is

$$\text{residual} = \text{observed} - \text{predicted.}$$

Here,

$$e_i = 160 - 166.46 = -6.46 \text{ cm,}$$

meaning that our model **overestimated** the height of that particular student.

- Section 8.2 of the textbook
- *Pearson's Correlation, Clearly Explained!!!* by StatQuest